

Lightning Solvers

Pehan Amarathunge

2 July 2024

1 Lightning Solvers

In order to solve Laplace's equation numerically in a bounded two-dimensional domain with corners, we will use a recently developed technique by Gopal and Trefethen[1] called *Lightning Solvers*. In this section, we will briefly explore the main mathematical ideas behind this method.

The lightning method is predominantly based on rational approximation theory and is essentially a generalisation of a key result in this field by Donald Newman[3]. A rational function is a ratio of two polynomials and we define a rational function of type (m, n) as a quotient of polynomials p/q , where p and q are polynomials of degree at most m and n respectively. We also define the order of a rational approximation as the maximum of the degrees of the numerator and denominator. In his 1964 paper, Newman considers the supremum norm approximation of $|x|$ on $[-1, 1]$ and proves that rational functions of order n , can achieve root exponential convergence (i.e. the error is $\mathcal{O}(\exp(-C\sqrt{n}))$ for some constant $C > 0$ as $n \rightarrow \infty$) as opposed to polynomials, which can only achieve $\mathcal{O}(n^{-1})$ convergence.

The result by Gopal and Trefethen extends this method to other singularities such as z^α or $z^\alpha \log(z)$, which are the standard types of singularities that are encountered when solving Laplace's equation in domains with corners. In particular Theorem 2.3 of [1] proves that by using rational functions we can approximate solutions to Laplace's equation in a convex polygon with corners with root-exponential accuracy. However, it is crucial that in order to attain this accuracy, the poles of the rational functions must be exponentially clustered near the corners.

Gopal and Trefethen suggested the following ansatz:

$$f(z) = \underbrace{\sum_{j=0}^{n_1} a_j (z - z_*)^j}_{\text{Runge Part}} + \underbrace{\sum_{j=1}^{n_2} \frac{b_j}{(z - z_j)}}_{\text{Newman Part}} \quad (1)$$

where z_* is a point near the centre of the domain (in fact in all applications in this paper we take z_* to be precisely at the centre of the domain) and z_j are a set of poles that are clustered exponentially close to the corners of the domain. The first sum is called the **Runge Part** and it corresponds to the smooth part of the solution. The second sum is called the **Newman Part** and it solves the singular part of the problem at the corners. As we will later discover, for certain geometries, the Runge part is only necessary in the ansatz.

In order to determine the sets of complex coefficients a_j and b_j , we will have to use the boundary condition on the walls and on the free surface. Since the boundary condition on the free surface is neither a Dirichlet nor Neuman boundary condition, but rather an eigenvalue equation, we will solve a generalised rectangular eigenvalue problem, in order to obtain these coefficients. This will be detailed in the subsequent section.

2 Generalised Rectangular Eigenvalue Problem

In order to determine the unknown set of coefficients, we will have to solve a rectangular generalized eigenvalue problem of the form

$$A\mathbf{x} = \lambda B\mathbf{x}. \quad (2)$$

Let Z be the $m \times 1$ vector of sample points on the boundary with the first m_1 points sampled on the free surface and let Z_1 be the $m_1 \times 1$ vector of sample points on the free surface. Then $A \in \mathbb{R}^{m \times (2n_1+2n_2+2)}$ represents the sampled value of the normal derivative of the basis functions in 1. On the other hand, $B \in \mathbb{R}^{m \times (2n_1+2n_2+2)}$ is a matrix, which has its first m_1 rows consisting of the value of the basis functions in 1 sampled on the free-surface and the rest of the matrix consists of zeroes. The vector $\mathbf{x} \in \mathbb{R}^{2n_1+2n_2+2}$ is the stacked vector of the real and imaginary components of the unknown coefficients. The matrices A and B are best understood as block matrices. Here we use MATLAB notation for succinctness. Pol refers to the set of poles that are clustered exponentially close to the boundary.

$$\mathbf{x} = \begin{bmatrix} \text{Re}[\mathbf{a}] \\ -\text{Im}[\mathbf{a}] \\ \text{Re}[\mathbf{b}] \\ -\text{Im}[\mathbf{b}] \end{bmatrix}, \quad A = \begin{bmatrix} R_n & N_n \end{bmatrix}, \quad B = \begin{bmatrix} R & N \\ \mathbf{0} \end{bmatrix}$$

where the blocks R and N correspond to the Runge and Newman parts respectively. Similarly, the blocks R_n , N_n correspond to the derivative of the Runge and Newman parts. These blocks are defined explicitly using MATLAB notation below:

$$R = \begin{bmatrix} \text{Re}(Z - z_*)^{(0:n_1)} & \text{Im}(Z - z_*)^{(0:n_1)} \end{bmatrix}, \quad (3)$$

$$N = \begin{bmatrix} \operatorname{Re}((Z - pol)^{-1}) & \operatorname{Im}((Z - pol)^{-1}) \end{bmatrix}, \quad (4)$$

$$R_n = \begin{bmatrix} \operatorname{Im}((0 : n_1)(Z - z_*)^{(0:n_1-1)}) & -\operatorname{Re}((0 : n_1)(Z - z_*)^{(0:n_1-1)}) \end{bmatrix}, \quad (5)$$

$$N_n = \begin{bmatrix} -\operatorname{Im}((Z - pol)^{-2}) & \operatorname{Re}((Z - pol)^{-2}) \end{bmatrix}. \quad (6)$$

Then it follows that $\operatorname{Im}[f'(Z)] = -\phi_y = A\mathbf{x}$ and for the first m_1 rows, $\operatorname{Re}[f(Z)] = \phi = B\mathbf{x}$. As a result, the first m_1 rows in the generalised eigenvalue problem in 2 enables us to determine coefficients that satisfy the eigenvalue boundary condition on the free surface¹ and the remaining rows deal with the no-normal flow condition on the rigid, impermeable walls.

In order to solve the rectangular generalised eigenvalue problem 2, we will employ a method similar to [2], that involves using the QR factorisation to 'square' the problem. We first perform a QR factorisation on A:

$$A = QR, \quad (7)$$

where $Q \in \mathbb{R}^{m \times n}$ is orthogonal and $R \in \mathbb{R}^{n \times n}$ is upper triangular. We then substitute 7 to 2 and then pre-multiply both sides by Q' (' denotes the transpose). We then arrive at the $n \times n$ square generalised eigenvalue problem:

$$Rx = \lambda Q'Bx, \quad (8)$$

which can be solved by calling the MATLAB routine `eig`.

References

- [1] Abinand Gopal and Lloyd N. Trefethen. "Solving Laplace Problems with Corner Singularities via Rational Functions". In: *SIAM J. Numer. Anal.* 57 (2019), pp. 2074–2094. URL: <https://api.semanticscholar.org/CorpusID:147704202>.
- [2] Behnam Hashemi, Yuji Nakatsukasa, and Lloyd Trefethen. "Rectangular eigenvalue problems". In: *Advances in Computational Mathematics* 48 (Nov. 2022). DOI: 10.1007/s10444-022-09994-8.
- [3] D. J. Newman. "Rational approximation to $|x|$." In: *Michigan Mathematical Journal* 11.1 (1964), pp. 11–14. DOI: 10.1307/mmj/1028999029. URL: <https://doi.org/10.1307/mmj/1028999029>.

¹In fact the generalised eigenvalue problem in 2 caters to the boundary condition $\phi_y = -\lambda^* \phi$, so in equation 2 we set $\lambda = -\lambda_*$